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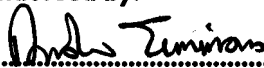
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METHODOLOGY FOR THE DEMONSTRATION OF COMPATIBILITY WITH 50 HERTZ SINGLE RAIL TRACK CIRCUITS

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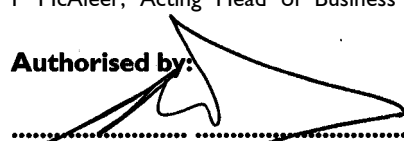
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1. Introduction

The purpose of this document is to provide a methodology for the demonstration of electromagnetic compatibility (EMC) of rolling stock with 50 Hz single rail track circuits installed on Railtrack controlled infrastructure. It is based upon previously accepted safety assessments which were undertaken for Traction and Rolling Stock thereby allowing them to operate over 50 Hz single rail track circuits.

The methodology is split into two distinct parts – one which considers the effect of steady state electromagnetic interference (EMI), and the other which considers the effect of transient EMI on the 50 Hz single rail track circuits.

2. Source Documents

2.1 Group Standards

- 1 Former BR Southern Region S&T Dept, Track Circuit Manual – Issue 02 dated 6/91

2.2 Line Standards

- 2 RT/E/PS/11757 Railtrack Line Product Specification "A.C. Phase Sensitive Track Circuits", Issue 1, dated December 2000
- 3 RT/E/S/11752 Railtrack Line Specification "Train Detection", Issue 1, dated December 2000

2.3 Previous Safety Case Documentation

- 4 "Class 365/5 Networker Express EE&CS EMC Safety Assessment for D.C. Operation, 50 Hz Single Rail Track Circuit Results of Traction Interference Modelling", AEA Technology Rail, 120-dr293, version a2, dated 11th December 1997.
- 5 "50 Hz Single Rail Track Circuit Model Validation", AEA Technology Rail, 120-dr0292, version a1, 7 December 1997.
- 6 "50 Hz Double Rail Track Circuit Traction Current Interference Susceptibility", AEA Technology Rail, 120-dr0262, version a0, 2 October 1997.

- 7 "Class 365/5 Networker Express EE&CS EMC Safety Assessment for D.C. Operation, 50 Hz Single Rail Track Circuit", AEA Technology Rail, I20-dr276, version b0, dated 18th December 1997.

2.4 Other Documentation

- 8 British Standard Institution: "BS EN 50121-3-1: 2000, Annex A: Calculation of the overall psophometric current of a train-set", Railway Applications – Electromagnetic Compatibility Part 3-1: Rolling Stock – Train and Complete Vehicle, December 2000.
- 9 "Final Report, Measurement of 50 Hz Harmonic Levels on 750V DC Traction Supply System for Railtrack PLC Contract No. 90218", Power Magnetics and Electronic Systems Limited, Log Ref: 2115, Issue 1, March 1996.
- 10 "Railway Safety - HM Chief Inspecting Officer of Railways' Annual Report on the safety record of the railways in Great Britain during 1995/6" HSE Books
11 "50 Hz Track Circuit Model - User Guide", AEA Technology Rail, ESID9165, Issue 10, January 2002

3. Description Of 50 HZ Single Rail Track Circuit

The normal arrangement on d.c. electrified lines is for both running rails to be used for the traction return current to provide the lowest possible resistance. This is the case on plain line and track circuits are therefore normally provided in a double rail configuration. However points and crossings and locations where the track length is short necessitate the use of single rail track circuits [1, 2].

Fig. 1 shows a typical single rail track circuit with the feed on the left-hand side of the diagram and the relay on the right. The signal rail is fitted with insulated rail joints (IRJs) at the track circuit boundary. The traction return rail is continuous and is not insulated from adjacent track circuits.

The principle of operation is when there is no vehicle within the track circuit's boundaries the current flows from the feed to the track circuit relay via one rail and returns by means of the other. This current lifts the relay vane, which in turn closes the front contacts of the relay. If a train occupies the track the wheels and axles constitute a low impedance path which connects the two rails together, effectively short circuiting the relay; current ceases to flow through the relay and consequently the vane drops, i.e. opening the front contacts.

The transformer is used to isolate the track circuit from the signalling supply. The circuit current is adjusted by means of the adjustable feed capacitor. The stabilising rectifier connected to the control coil of the relay is intended to limit the effect of interference. This device is a selenium rectifier which presents a low impedance at high voltages, such devices can fail and an open circuit failure is not revealed. Hence to ensure that the analysis is pessimistic it is prudent to assume that this device has failed i.e. open circuit. [4].

The relays used in 50 Hz track circuits are double element vane relays, which contain two coils. One coil, called the local coil, is connected to the 50 Hz signalling supply, whilst the second coil, denoted the control coil, is connected to the signal rail and traction return rail [5].

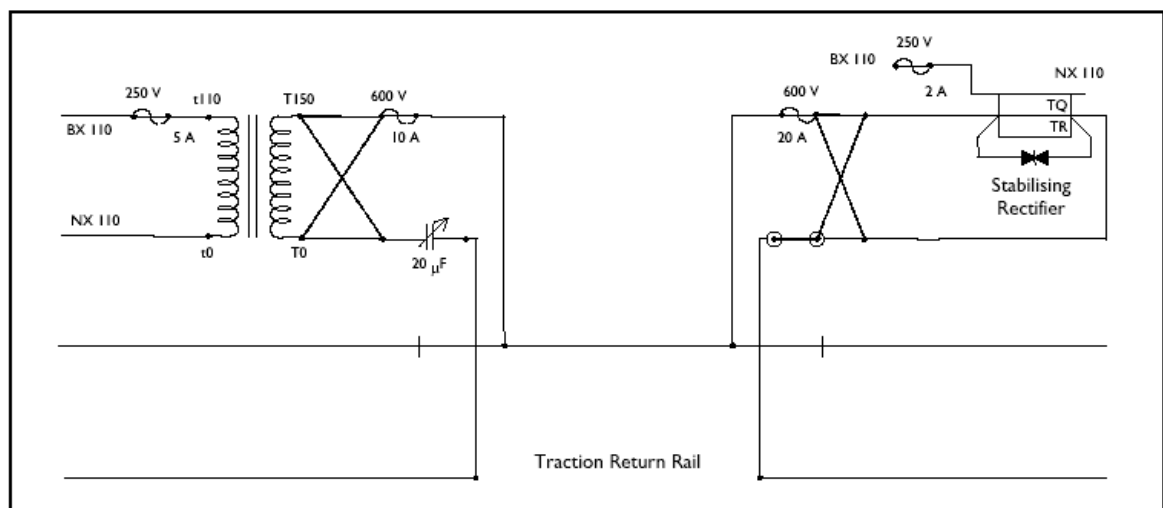


Figure 1: 50 Hz single rail track circuit configuration.

The maximum length of single rail track circuits is normally 200 m, but may increase to 300 m in terminal platforms. As a general principle track circuits are arranged so that trains normally enter at the feed end and exit at the relay end [1].

There are three types of relay in common use: VT1 (and the VT1(SP) variant), CE391 and G4 of which the VT1 relay is the most common [6]. Each of the relay types has its own characteristic with different control coil impedance, pick up voltage and drop away voltage (table I). The control coil impedance is frequency dependent and the value shown in table I refers to the nominal value at a frequency of 50 Hz [6].

Relay type	50 Hz impedance / Ω	Pick-up voltage / V	Drop away voltage / V
VTI	$3.5 + j\ 0.2$	0.50	0.45
CE39I	$5.2 + j\ 2.5$	1.16	0.74
G4	$1.7 + j\ 4.0$	0.60	0.57

Table 1: Relay Parameters.

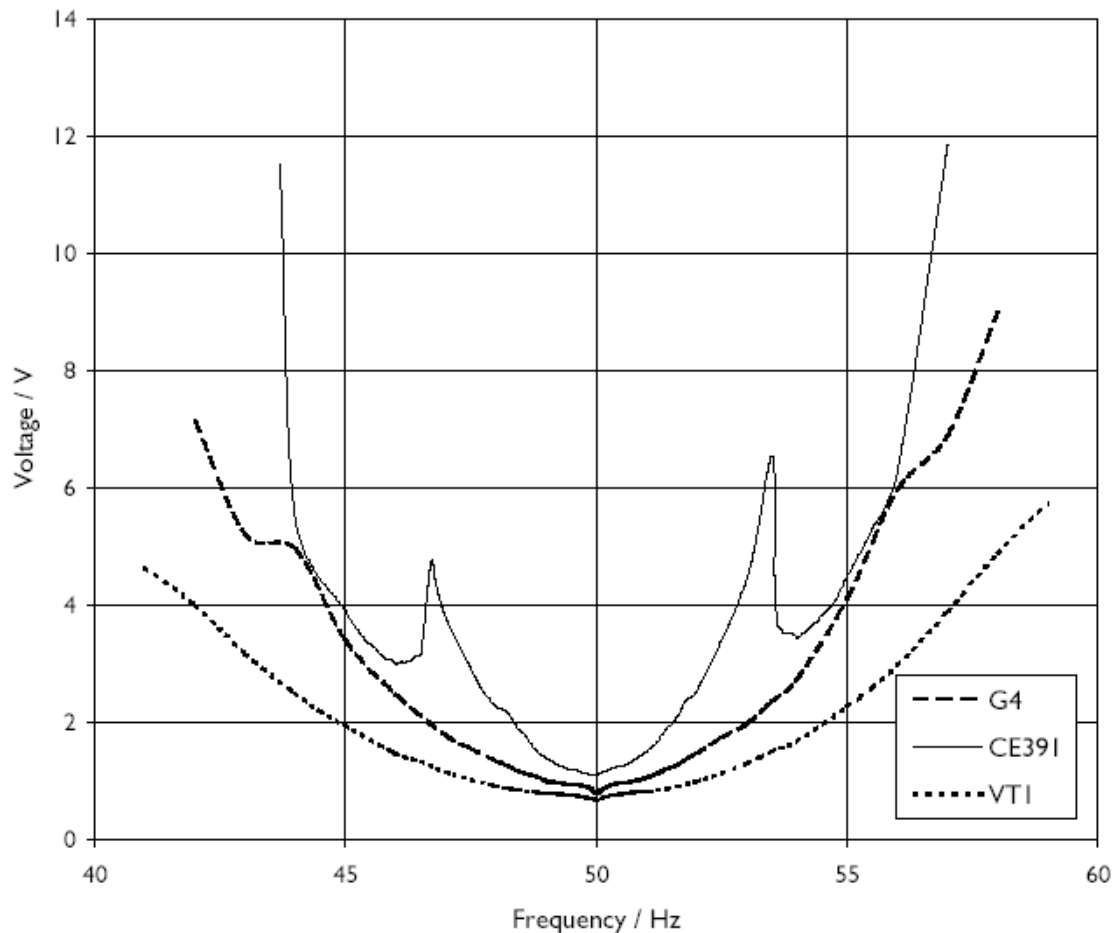


Figure 2: 50Hz Relay responses (pick up voltage) for a control coil frequency of 50 Hz

N.B. This figure does not imply that the relay can be adequately represented by a filter characteristic

Typical responses for the three common relay types are shown in figure 2. This figure shows the voltage required to 'pick up' the relay for frequencies around 50 Hz. The lowest point of the response curve corresponds to the frequency of the control coil and is therefore dependent on the frequency supplied to this coil. The only situation that will result in the relay being picked indefinitely is when the control and local coils

are supplied with the same frequency and the phase relationship is correct. Any other frequency will result in an oscillating torque (at the beat frequency) being induced in the vane and the relay will alternately pick and drop. Therefore a filter representation of the response shown in figure 2 is not appropriate for the modelling of 50 Hz track circuits.

4. Infrastructure Assumptions And Terminology

The methodology presented utilises the following assumptions:

- 50 Hz single rail track circuits are correctly installed and maintained in accordance with the Track Circuit Manual and/or the relevant Railtrack Company Product Specification [1, 2]. Whilst the Track Circuit Manual is a superseded document it is considered an adequate reference source for 50 Hz track circuits as it was the installation standard in use when the majority of them were installed.
- The length for assessment of the 50 Hz single rail track circuit is assumed to be 200 m.
- Track circuits with a length greater than 200 m are assumed to have cross bonding which means that not all the traction return current from a train will flow the full length of the track circuit. Track circuits longer than 300 m are generally installed in terminal platforms where there is likely to be more cross bonding and fewer trains will be able to draw traction current over the track circuit in question.
- No 'slow to pick' track repeat relays (TPR) are fitted. It is known that not all track circuits are fitted with TPR's and of those that are, it cannot be assumed that only the TPR contacts are used for signalling/control functions.
- To consider a normal (unfailed) pessimistic track circuit, all traction current is assumed to flow via the traction rail in one direction.
- Absence of the Westalite stabilising rectifier
- The term 'new train' refers in this document to a train subject to EMI analysis and to be introduced on routes with 50 Hz single rail track circuits.
- The term 'existing train' refers in this document to a train currently operating on the route of interest. It is assumed that existing trains operate with an acceptable level of safety on 50 Hz single rail track circuits fitted on the route [1], [7].
- The track is assumed to be of standard gauge
- The sleepers in use are assumed to be concrete or wood, steel sleepers are not included in the analysis presented here.
- UIC 60 Rail is not included in the assessment as presented. The methodology given here is applicable but the parameters of this rail type at the frequency of interest will need to be determined and used in the appropriate calculation.

- The presence of check rails will reduce the effective impedance of the rail to which they are attached. This is not included in this document, however the methodology provided allows for the effect of such differences to be evaluated.

5. Susceptibility Analysis

Due to the dynamic characteristic of the track circuit relay, the effects of steady state EMI are quite different than that from transient EMI. Where steady state and transient can be defined as:

- **Steady state** A particular level of current which exists for a time greater than the general range of relay operating times. Typically this is a time greater than 0.5 seconds. For steady state analysis only the amplitude of the current is of interest.
- **Transient** Short duration and/or changing levels of current when compared to the dynamic response of the relay vane, typical events being step changes in traction current. The response of the relay will lag that of the electrical initiating event and may occur after the electrical event has ceased. For this analysis the mechanical dynamic performance of the relay must be included in the analysis.

5.1 Substation Voltage Ripple

It is known that the d.c. traction power supply contains a 50Hz component due to its derivation from the mains supply. Due to this voltage ripple on the supply any train will conduct a 50 Hz current dependent on the impedance that the train presents to the supply at 50 Hz. Therefore the 50 Hz current due to a train present on a particular track circuit is made up of two components, that due to conduction of 50 Hz current from the supply and that generated by the train itself.

Measurements have indicated that the 50Hz harmonic ripple on the d.c. traction supply is of the order of 0.1 V to 2 V [9]. However the level of 50 Hz ripple to be used in any particular safety assessment should be declared to Railtrack at the start of the project.

5.2 Steady state analysis

Since the electrical supply to the local coil of the relay is ultimately derived from the mains then the frequencies of interest generated or conducted by the train are those between 48 Hz and 52 Hz thereby allowing for the effect of mains frequency variation

on the local coil. The track circuit being a vane relay, which is an electromechanical device, has a dynamic response characteristic. This precludes it being modelled simply as a filter; differences between the frequencies of the currents flowing in the local and control coils can result in a beating torque being set up in the vane. Generally these are shorter than the response time of the vane and it does not react to them. The vane relay will only respond to low frequency torque oscillations.

5.2.1 Interference Mechanism

The configuration of the interference mechanism for single rail 50 Hz track circuits is shown in fig. 3. The equivalent circuit of the interference mechanism is shown in fig. 3. The axle indicates the location of the train.

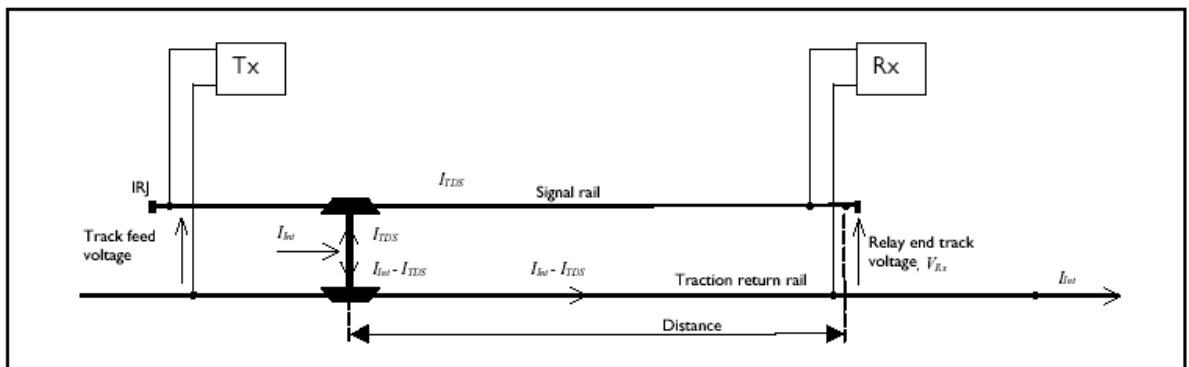


Figure 3: 50 Hz track circuit configuration.

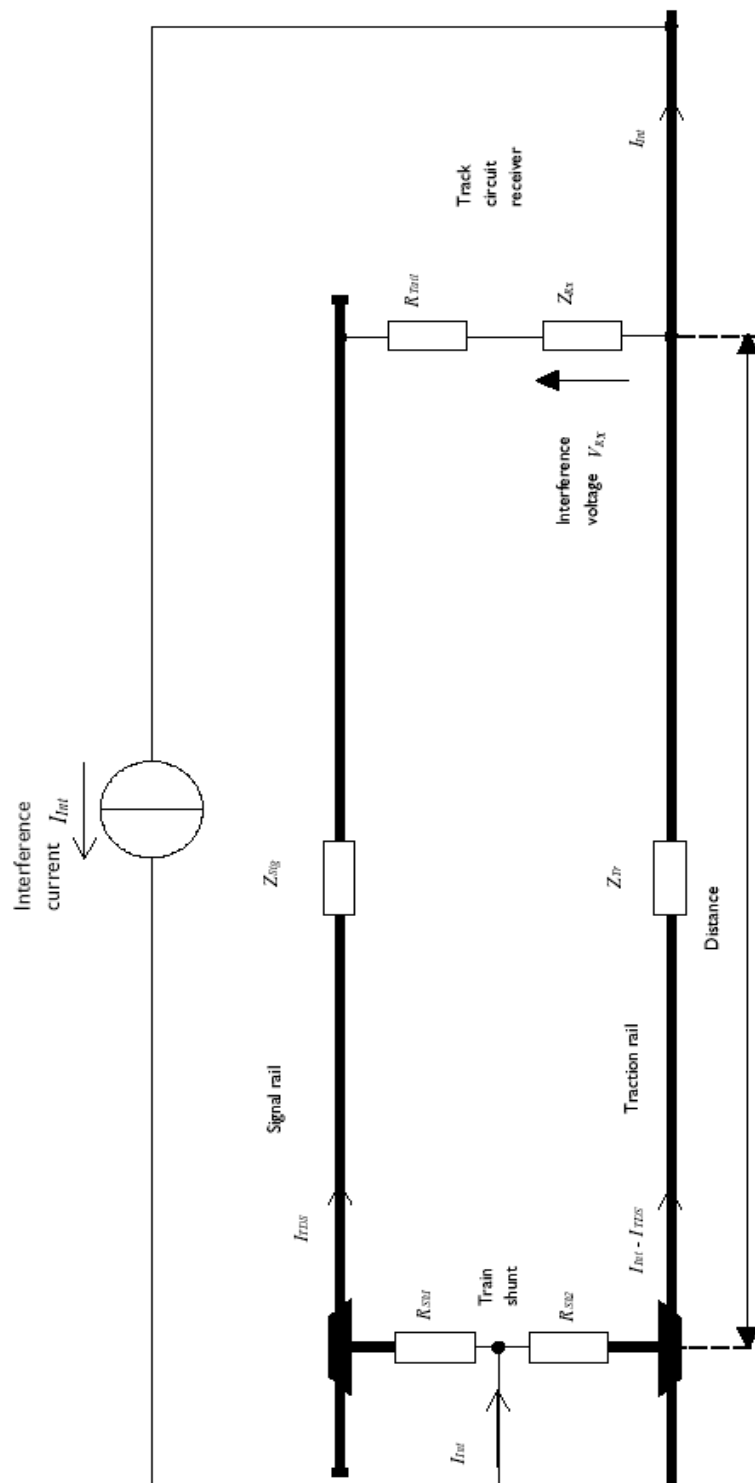


Figure 4: Wrong side failure mechanism.

Where

IRJ	Insulated Rail Joint
I_{TDS}	Current flowing in the train detection circuit
I_{Int}	Interference component of traction return current
V_{Rx}	Voltage across track circuit receiver
R_{Sh1}	Train shunt resistance
R_{Sh2}	Train shunt resistance
R_{Tail}	Resistance of track circuit tail cables
Z_{Sig}	Per unit impedance of signal rail
Z_{Sig}	Impedance of signal rail
Z_{Tr}	Per unit impedance of traction return rail
Z_{Tr}	Impedance of traction return rail

Equation 5.1 gives the interference current limit as a function of the voltage across the relay based on fig. 4. If the interference voltage V_{Rx} is of sufficient magnitude the track circuit relay fails to drop away. From knowledge of the relay drop away voltage an interference current limit can be calculated from equation. 5.1.

$$I_{Int} = \left(\frac{R_{sh1} + R_{sh2} + Z_{Tr} + Z_{Sig} + R_{Tail} + Z_{Rx}}{(R_{sh2} + Z_{Tr}) \cdot Z_{Rx}} \right) \cdot V_{Rx} \quad (5.1)$$

If R_{Sh1} and R_{Sh2} are neglected (a reasonable assumption since the train shunt is very small), the interference current limit I_{int} increases with decreasing Z_{Tr} . Additional bonding of the traction rail would result in a decrease in Z_{Tr} thus the worst case to be considered is when there is no cross bonding of the traction return rail and Z_{Tr} is at a maximum.

Since $Z_{Tr} = Z'_{Tr} \times (distance)$ and $_{Sig}Z = Z'_{Sig} \times (distance)$ equation. 5.1 can be rewritten (equation. 5.2).

$$I_{Int} = \left(\frac{R_{sh1} + R_{sh2} + Z'_{Tr} \times (distance) + Z'_{Sig} \times (distance) + R_{Tail} + Z_{Rx}}{(R_{sh2} + Z'_{Tr} \times (distance)) \cdot Z_{Rx}} \right) \times V_{Rx} \quad (5.2)$$

Neglecting R_{Sh1} , and R_{Sh2} equation. 5.2 can be simplified to equation. 5.3.

$$I_{Int} = \left(\frac{1}{Z_{Rx}} + \frac{Z'_{Sig}}{Z'_{Tr} \times Z_{Rx}} + \frac{1}{Z'_{Tr} \times (distance)} + \frac{R_{Tail}}{Z'_{Tr} \times (distance)} \right) \times V_{Rx} \quad (5.3)$$

If $Z_{Rx} \gg Z'_{Tr} \times (distance)$ and $Z'_{Sig} \times (distance)$

And given $Z'_{Sig} = Z'_{Tr}$

$$\text{Then } I_{Int} = \left(\left(\frac{1}{Z'_{Tr} \times (distance)} \right) \left(1 + \frac{R_{RTail}}{Z_{Rx}} \right) \right) \times V_{Rx} \quad (5.4)$$

From equation 5.4 it can be seen that increasing distance results in decreasing I_{Int} .

Since for the determination of the steady state interference limit the worst case must be considered, analysis should be carried out with the train located at the feed end of the track circuit, which represents the maximum distance.

5.2.2 EMI budget

The interference current limit I_{Int} derived in the previous section gives the total allowable level in a fault free track circuit. The allowable level for a single train must be lower, since some allowance must be made for other than trains and other interference sources, such as track feeds. The single train current limit represents the total 50 Hz current due to the train, including both the current conducted from the supply and that generated by the train itself.

For normal (fault free) track circuits, the total allowable interference current is divided into three equal portions. One portion represents the allowance for the train under analysis, one represents other trains and one acts as a safety margin [6]. When the interference mechanism only exists for a limited time, such as the time taken to detect and repair infrastructure, then an increased allowance for train current is appropriate. In these cases it is assumed that the train can produce up to half the maximum permissible interference current. The other half being allocated to interference from other sources.

The relay drop away voltage is used to determine susceptibility since its use overcomes the risk of delayed drop away, this could occur if a train entered the section under consideration and the interference emanating from it was sufficient to keep the relay energised. Therefore V_{Rx} is the relay drop away voltage.

5.2.3 Steady state train data processing

The manufacturer is to establish the maximum steady state current for a train in the

following conditions

- Fully functional and without faults
- Functional but in degraded mode i.e. various sub systems shut down or isolated
- Functional but with faults present

The above conditions are to be applied to all combinations of train formation, equipment configuration and operating conditions. Auxiliary systems as well as traction drive systems must be considered.

The frequency range to be considered is 48 Hz to 52 Hz. Where trains are made up of multiple systems i.e. coupling units together then a single point measurement may not be practical. In which event the total current of a train shall be derived by adding arithmetically the currents from all sources, where the sources are synchronised or where such a mechanism could exist. If it can be demonstrated that the sources are not locked together and therefore cannot appear in phase for long enough to activate the vane relay e.g. less than 250 ms then the RSS (Root Sum Squared) method may be used. Further guidance may be found in ENV50121-3-1 [8].

5.2.4 Comparison of train interference level with steady state limit

If the total train interference current is determined as described in 5.2.3 and it is below the calculated interference limit, then the train is compliant with the steady state requirements for 50 Hz single rail track circuits.

5.2.5 Example

5.2.5.1 Base Case For Normal 50 Hz Single Rail Track Circuit

Given: A track circuit configured as shown in fig. 4, where

$$\begin{aligned}
 V_{Rx} &= 0.45 \text{ V} \\
 Z_{Rx} &= 3.36 \, \Omega + j \, 0.228 \, \Omega \\
 \text{Distance} &= 200 \text{ m} \\
 Z_{Sig} = Z_{Tr} &= 1.694 \, 10^{-4} + j \, 3.48 \, 10^{-4} \, \Omega/\text{m} \\
 R_{Sh} = R_{Sh1} = R_{Sh2} &= 100 \, \mu\Omega, \\
 R_{Tail} &= 0
 \end{aligned}$$

Calculation of total interference current limit using equation 5.2

$$\begin{aligned}
I_{Int} &= \left(\frac{R_{sh1} + R_{sh2} + Z'_{Tr} \times (\text{distance}) + Z'_{Sig} \times (\text{distance}) + Z'_{Rx}}{(R_{sh2} + Z'_{Tr} \times (\text{distance})) \times Z'_{Rx}} \right) \times V_{Rx} \\
&= \left(\frac{200 \times 10^{-6} + 2 \cdot (1.694 + j \cdot 3.48) \cdot 10^{-4} \cdot 200 + (3.36 + j \cdot 0.228)}{(100 \times 10^{-6} + (1.694 + j \cdot 3.48) \cdot 10^{-4} \cdot 200) \cdot (3.36 + j \cdot 0.228)} \right) \times 0.45V \quad (5.6) \\
&= (2.8156 - j5.2391)A
\end{aligned}$$

The rms value of the total interference current limit is $I_{Int} = 5.95$ A. Application of EMI budget gives a single train interference current limit of $\frac{I_{Int}}{3} = 1.98$ A.

5.2.6 Failed Infrastructure

With a broken rail situation the track circuit susceptibility is increased considerably. Equation 5.4 shows that the allowable interference current reduces as the traction rail resistance increases. With typical cross bond spacing the effective increase in traction return current length is such that the allowable 50 Hz currents would be unrealistically low. Consequently it may be difficult to demonstrate compatibility by calculation. In these circumstances one approach is to demonstrate that the steady state levels for the new train are no greater than that produced by stock currently running over the route. The objective being to demonstrate that the introduction of the new trains will not increase the risk above that which already exists.

If a calculation is required then the circuit diagram, equivalent circuit and formula are as follows:

For the track circuit configured as in figure 3 with a broken rail the traction return current is assumed to flow via the cross bond and then via adjacent track thereby increasing circuit impedance and susceptibility.

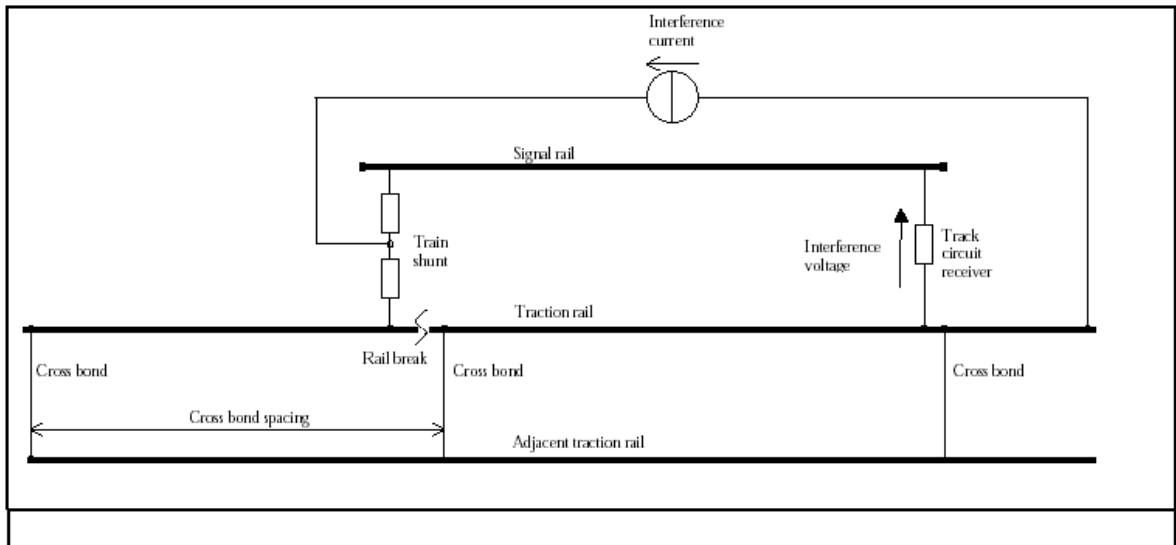


Figure 5: Broken rail configuration

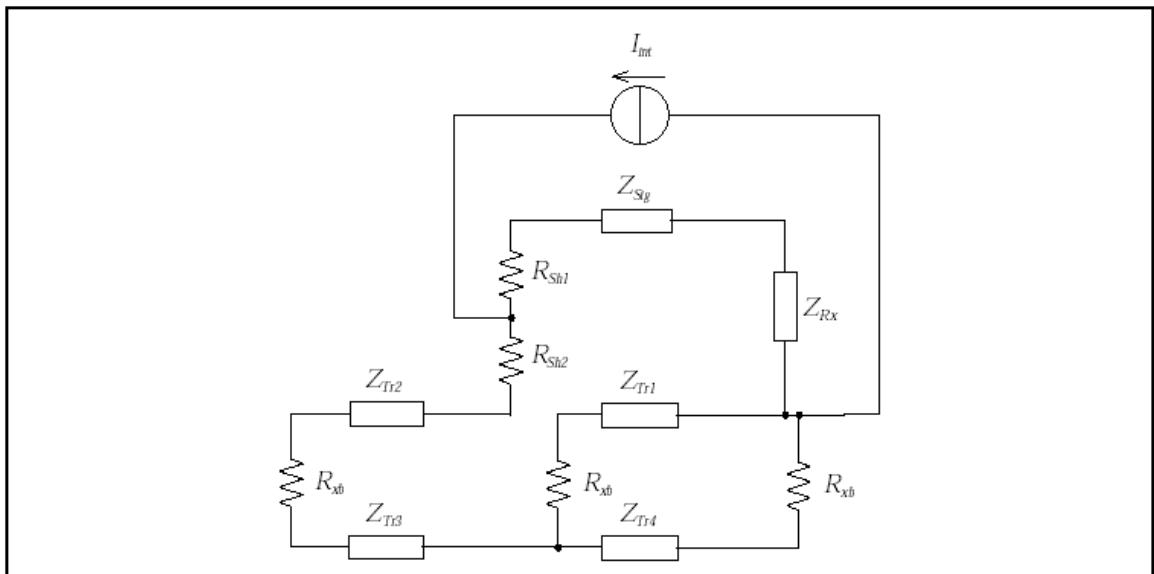


Figure 6: Electrical representation of faulty infrastructure scenario.

Fig. 6 is the electrical circuit representation of fig. 5. The formula (5.7) is used to determine I_{Int} for a given voltage V_{Rx} at the track circuit relay.

$$I_{Int} = \frac{R_{xb} + R_{sh} + 2 \cdot Z_{Tr} + 0.5 \cdot (R_{xb} + Z_{Tr}) + R_{sh} + Z_{Tr} + Z_{Rx}}{Z_{Rx} \cdot (R_{xb} + R_{sh} + 2 \cdot Z_{Tr} + 0.5 \cdot (R_{xb} + Z_{Tr}))} \cdot V_{Rx} \quad (5.7)$$

Where

I_{Int}	Interference component of traction return current
V_{Rx}	Voltage across track circuit receiver
R_{Sh1}	Train shunt resistance
R_{Sh2}	Train shunt resistance
Z_{sig}	Impedance of signal rail
Z_{Tr}	mpedance or traction return rail where $Z_{Tr1}=Z_{Tr2}=Z_{Tr3}=Z_{Tr4}=Z_{Tr}$
R_{xb}	Cross bond impedance

If absolute compatibility cannot be demonstrated with the broken rail scenario, an alternative would be to demonstrate that the probability of the train exceeding this limit coincident with the increased susceptibility due to the infrastructure fault is acceptably small.

5.3 Transient analysis

Due to the dynamic characteristic of the track circuit relay, its' response depends not only on the interference level but also on the rate of change of traction current and the phase relation between relay local coil voltage and relay control coil voltage during these events. Therefore a simple maximum value of interference current/time can not be given. It is therefore recommended that a comparative analysis between the new trains and existing trains already operating over the route is undertaken. The purpose of this comparison is to demonstrate that the introduction of the new trains on the route will not increase the level of existing risk.

The methodology employed to make this comparison is that the transient performance of both the new and the existing trains is measured, recorded and then played through a model of the track circuit. The model's reaction to the transients from each of the trains can then be observed and compared, then an assessment made. It is important to note that the train transients used must include all those expected from normal and degraded operations as well as the effects of faults. The latter condition may use the predicted effect of faults as it is not always practical to replicate or create an actual fault condition. Furthermore it may not be possible to compare the effects of faults between different designs therefore a probabilistic assessment to determine the frequency of faults will be a necessary and fundamental part of the assessment.

Railtrack possesses a validated model of the single rail 50 Hz track circuit that is suitable for the use in the production of rolling stock EMC safety cases. This model can be provided to persons presenting such a case. Detail on the requirements and use of this model can be found in the user guide for the model [11]. The methodology for building and validating a model is given in this section.

5.3.1 The track circuit model

In contrast to the steady state analysis, where only the 50 Hz transfer impedance between the interference current and relay voltage is of interest, the transient analysis must also take the relay's dynamic characteristic over a range of frequencies. For analysis purposes, the track circuit is divided into its principal constituent parts, which are the track circuit and the relay. Therefore the numerical track circuit model can be structured as shown in fig 7.

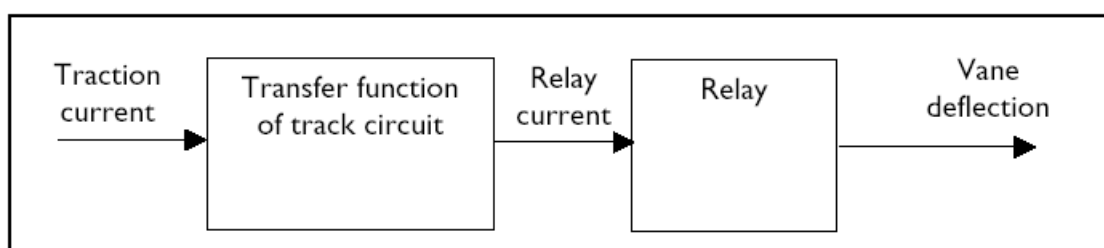


Figure 7: Structure of the track circuit model.

Subsequent sections detail the methodology to be employed in constructing a model but it should be recognised the purpose of the model in this instance is to allow a comparison to be made about the effect of transient current waveforms from two trains. Consequently whilst good correlation between the performance of the model and an actual relay to transients is required it is not the intention that the performance of the 50 Hz track circuit be replicated precisely in every respect. However where compromises are made, then the model should give a pessimistic result when compared to an actual track circuit.

5.3.1.1 Description and application of the track circuit transfer function

Whilst for the steady state analysis only the 50 Hz impedance is of interest, the transient analysis must consider a wider range of frequencies. The frequency dependent transfer impedance between the interference current and the relay voltage is given equation 5.8 and is derived from fig. 3.

$$V_{Rx}(f) = \frac{Z_{Tr}(f) \cdot Z_{Rx}(f)}{Z_{Tr}(f) + Z_{Sig}(f) + R_{Tail} + Z_{Rx}(f)} \cdot I_{Int}(f) \quad (5.8)$$

$$I_{Rx}(f) = \frac{Z_{Tr}(f)}{Z_{Tr}(f) + Z_{Sig}(f) + R_{Tail} + Z_{Rx}(f)} \cdot I_{Int}(f) \quad (5.9)$$

$$\text{Where} \quad I_{Rx}(f) = \frac{V_{Rx}(f)}{Z_{Rx}(f)} \quad (5.10)$$

Variation of rail impedance with frequency is a complex characteristic due primarily to the “skin effect” which affects the impedance from as low a frequency as 12 Hz. Appendix F provides information on rail resistance.

5.3.1.2 Calculation of the relay control coil current and voltage

The relationship between the interference current and the relay control coil current is established, equation 5.9. The approach is to derive the relay control current from measured train data by use of equation 5.9. Therefore equation 5.9 is applied to train current data recorded with a current acquisition system which has a minimum sampling rate of 2 kHz.

One approach is to process the data by use of a windowed Fourier Transform process, alternatively a digital filter implementation of equation 5.9 may be used. Where the Fourier transform approach is adopted, care should be taken to remove any windowing effects from the process.

5.3.2 Description and application of the relay model

The basic a.c. vane relay contains an aluminium vane, mounted on a shaft, which is mechanically linked to a number of movable relay contacts. The vane can freely move between two mechanical stops: the lower, or “back”, stop and the upper, or “front”, stop. The linkage is designed such that when the vane sits on the lower stop, the back contacts are closed, and when the vane sits on the upper stop the front contacts are closed. Two laminated iron cores are positioned around the vane, and on these the relay coils are wound, see fig. 8. There are two main coils: the local coil, which is supplied from the 110 V a.c. signalling supply, and the control coil, which is supplied from the track circuit. In the VTI there is also a tertiary coil, which is a second control core coil used to tune the effective control coil impedance to 50 Hz. The currents in these coils produce magnetic fluxes which link the vane and induce eddy currents within it. Interaction of the eddy currents and fluxes can produce a torque on the vane which is sufficient to lift it and thereby close the front contacts.

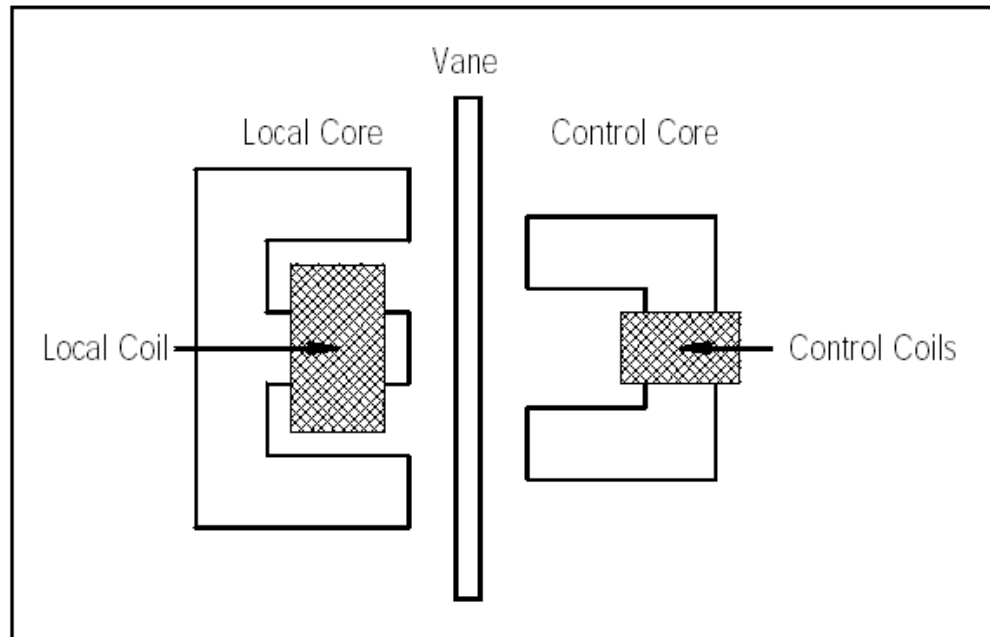


Figure 8: Relay core arrangement.

5.3.2.1 The developed torque

Equation. 5.11 describes the torque as function of the local and control coil fluxes [7]. As in any electromechanical device, the basic mechanism of force generation is defined by:

$$T = k \cdot \left(\phi_c(t) \cdot \frac{\partial \phi_l(t)}{\partial t} - \phi_l(t) \cdot \frac{\partial \phi_c(t)}{\partial t} \right) \quad (5.11)$$

Where

- $\phi_c(t)$ = the instantaneous flux produced by the control coil
 $= i_c(t) \times k_c$
 where k_c is a constant related to the construction of the control coil
- $\phi_l(t)$ = the instantaneous flux produced by the local coil
 $= i_l(t) \times k_l$
 where k_l is a constant related to the construction of the local coil
- k = constant related to the vane size and core size

5.3.2.2 Restoring Torque

For transient conditions the response of the vane is not directly related to the restoring torque since three reaction forces, inertia, frictional and restoring act on the vane. At any instant any one of these may dominate the response. In addition account should also be taken of the stops and the spring contact forces associated with them.

5.3.2.3 Model Block Diagram

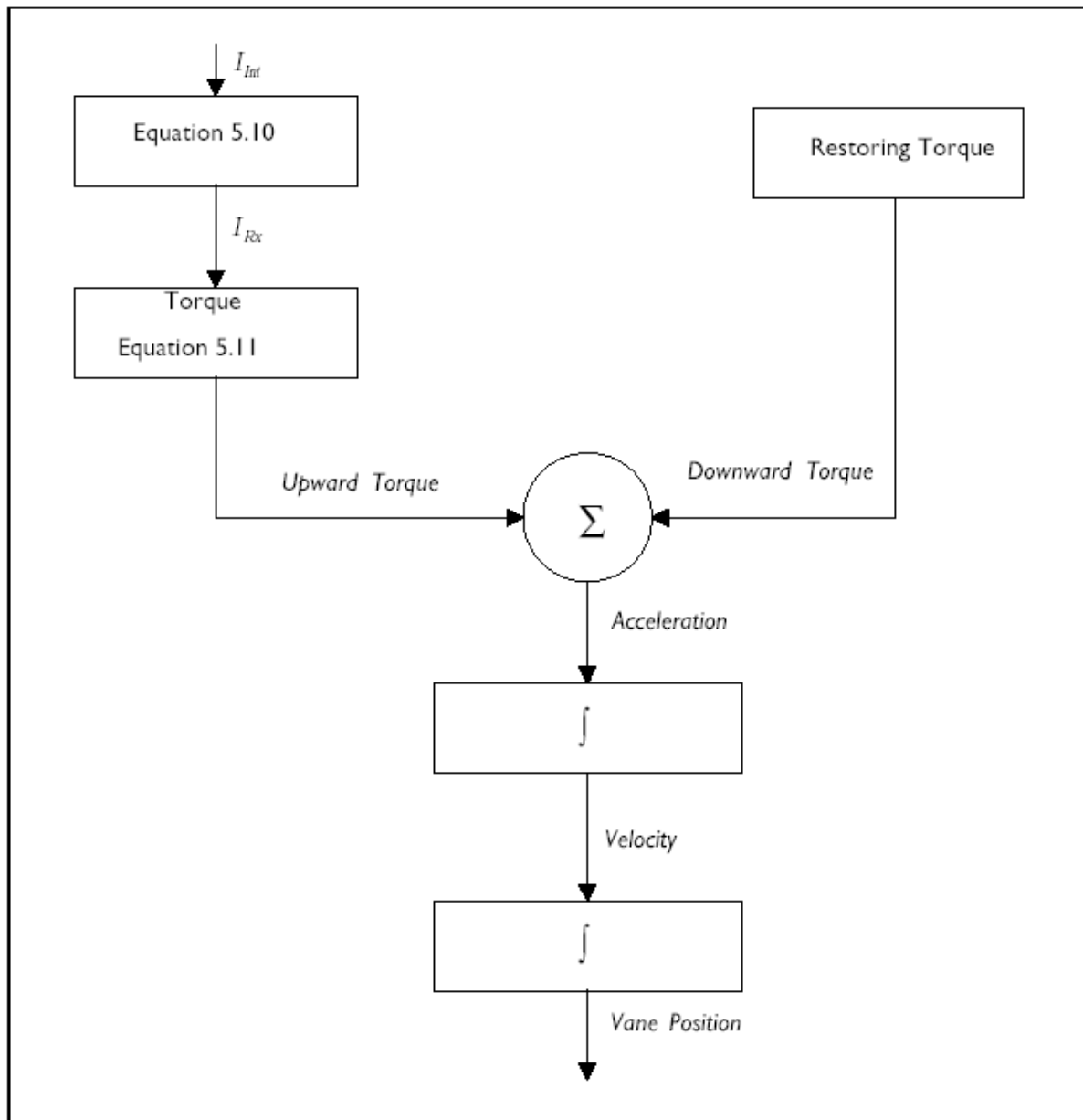


Figure 9: Relay Functional Elements

5.4 Validation

5.4.1 Relay model validation

The derived relay model must be validated before it is used to assess the effects of EMI generated by trains on 50 Hz track circuits.

5.4.1.1 Steady state validation

The steady state performance is validated by comparing the model's response to a given input with that of a real relay for the same input which is used as a reference.

To produce a reference response, the local coil of a relay is connected to a 110 V / 50 Hz supply, a 50 Hz voltage displaced by 90° from that connected to the local coil is applied to the control coil. The control coil voltage is varied in magnitude and the relay's pick up point is recorded. Since the test reflects steady state condition, the control coil voltage must remain unchanged for a duration exceeding the dynamic time constant of the relay by at least a factor of 5. It is expected, that keeping the control coil voltage unchanged for a duration of 1 s fulfils this criterion [5]. By recording the relay contact state versus the time for each control coil voltage, the reference response of the relay is established.

The experimental tests carried out are repeated accordingly on the model. The state of the relay contacts is recorded versus the time and compared with the reference response. If the model response matches the reference response closely then the steady state accuracy of the model is acceptable.

5.4.1.2 Transient validation

Validation of the transient response is carried out by comparing the vane movement of the model with the vane movement of a real relay. Reference waveforms showing the contact state versus time are recorded while the relay is subjected to a number of transient control coil voltages and the local coil is supplied by 110 V / 50 Hz. Since the phase between local coil and control coil voltages must be taken into account, these voltages have to be recorded simultaneously during the tests. The recorded local and control coil voltages are applied to the relay model. After completing the simulations the vane movements and contact state time curves of the relay model are compared with the reference curves.

It is suggested that the model is validated using standard signals and severe transient signals as transient control coil voltages [5]. Standard signals should include step changes and ramps varying in magnitude and/or rise time.

5.4.2 Track Model Validation

The parameters used in the track model should be checked to ensure that they conform with those given in the Track circuit handbook, table 2.

Track Circuit Handbook – Rail Impedance	
$f \text{ Hz}$	$Z_{\text{Rail}}(f) \Omega \text{ km}^{-1}$
0	0.035
50	0.3
400	2.5
2000	10

Table 2: Rail impedance data

Validation of the multi-frequency transfer function is best accomplished by a comparison of measured against simulated control currents for a given traction data input data sample containing representative samples – a data length of approximately 60 seconds duration is recommended.

6. ANALYSIS USING MODEL

6.1 Simulation Method

To implement the track circuit model a suitable mathematical software package capable of processing large amounts of data is required.

To demonstrate that the EMI of the new train does not increase the risk of wrong side failure of 50 Hz single rail track circuits data is recorded both from the existing trains and the new train operating over the same route. Both sets of data are then played through the track circuit model. To ensure that the due account is taken of the phase angle relationship between the local and control coil each section of data needs to be played through 8 times, on each occasion the phase angle should be increased by 45 °.

The data to be used for the analysis must contain the transients generated by the events listed in table 3 (section 9). During each simulation of the new train and the existing train, the events listed in table 3 are to be recorded and analysed in such a manner so as to allow meaningful comparisons between trains to be made.

The test data is applied to the track circuit transfer function (section 6.2.1.2) thereby determining the relay current. In processing the data the following may be taken into account;

- It can be expected, that the amount of traction current data is beyond the processing capabilities of conventional data analysis software. It might therefore be necessary to split up the data in windows each containing a power of 2 number of samples, e.g. 2¹⁷ samples. Using a base 2 number of samples allows the application of the Fast Fourier Transform (FFT) to the current data.
- Each data window is transformed into the frequency domain using the FFT method.
- The relay current is calculated for each frequency component using equation 5.9. Look-up tables provide the rail and relay frequency characteristic required for equation 5.9. Whilst the look up table describing the rail frequency characteristic was calculated as described in Appendix F section F1, the look up table for the relay is based on measurements (Appendix F section F2).
- The inverse FFT is applied to the calculated relay current in order to obtain the relay current in the time domain.
- The end effects occurring at the boundaries of the data windows are removed. Alternatively the non-linearity with frequency can be built into the track circuit model and the raw current signal applied. A simplified model representing the impedance of the control coil is sufficient to enable the resulting relay current to be derived. The relay current when applied to a relay model will allow the vane position and the state of the relay contacts to be calculated.

6.2 Evaluation of the Simulation Results.

The vane deflection resulting from the analysis of the data gathered from both the new and the existing trains is to be analysed. The results of the analysis (vane deflection and duration) are then plotted graphically.

In the analysis process the number of deflections of the vane above a given level is plotted against that level, such that the graph shows the number, or rate, of vane movement on the y-axis against the level of movement on the x-axis.

To obtain the duration of vane deflection above a certain level, the time the vane deflection exceeds this level has to be calculated for each transient. Denoting this time t_{ex} , then the proportion of the testing time the vane moves beyond the x-axis level, can be described by equation 6.12, where t_{Test} expresses the time span of the test.

$$\text{Relative duration of vane deflection} = \frac{t_{ex}}{t_{Test}} \quad (6.12)$$

The compatibility is based on the vane deflection curve representing the new train compared to a curve representing the existing train. Fig. 10 and fig. 11 are example plots demonstrating how the rate of vane deflection is plotted versus the exceedance level.

The important criterion is the time spent at each vane deflection relative to the total time of the test run for each of the trains under test. A comparison of the absolute number of vane deflection events is not an important criterion.

In fig. 10, the relative duration of vane deflection of the new train is lower than the duration of vane deflection of the existing train. This means that the EMI emanating from the new train will have a lesser effect on 50 Hz single rail track circuits than the existing trains.

For the example shown in fig. 11 the result and the conclusion is not as obvious as that in fig. 10, some vane deflections resulting from the new train having a longer duration than those from the existing trains. However if the magnitude of vane deflection is considered in conjunction with their relative duration then it can be seen that the new train has a higher proportion of vane movements with smaller deflections than the existing train where the reverse is true. Thus the EMI from the new train results in less severe vane movements consequently the proportion resulting in front contact closure is lower than with the existing train. It can therefore be concluded that the new train performs better than the existing train with respect to 50 Hz single rail track circuits.

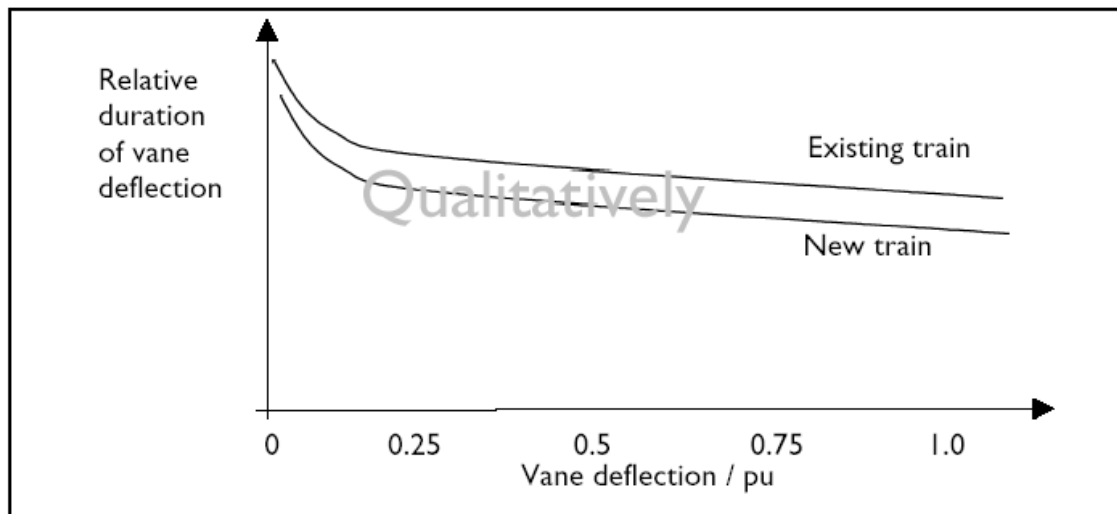


Figure 10: Typical test results (1)

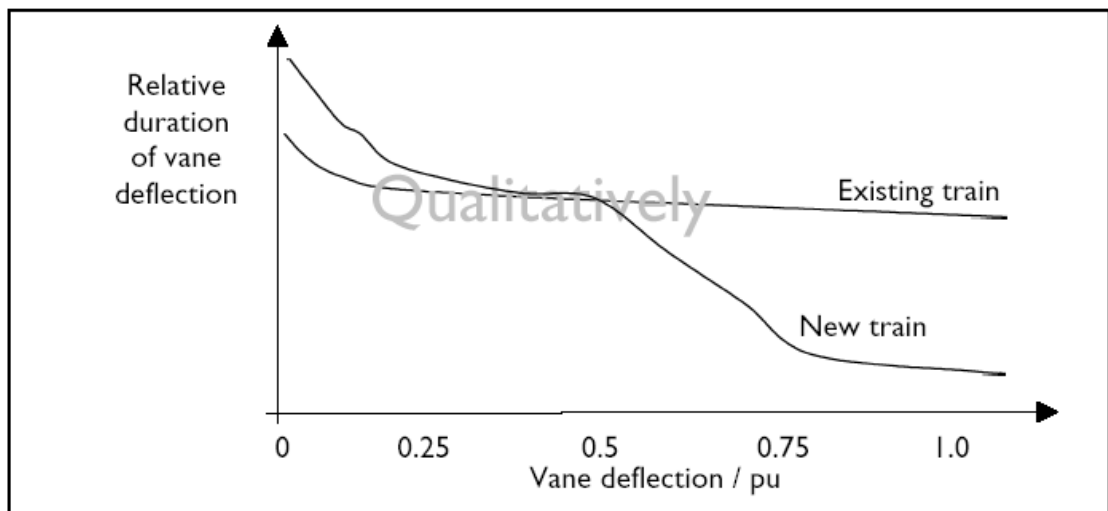


Figure 11: Typical test results (2)

7 Axle To Axle Voltage – Example

For a single rail track circuit, occupied by a train, a wrong side failure may be caused by a train on an adjacent track producing an axle-to-axle voltage. Figure 12 shows how a train on track 2 can cause a current to flow and hence energise the track circuit relay on track 1.

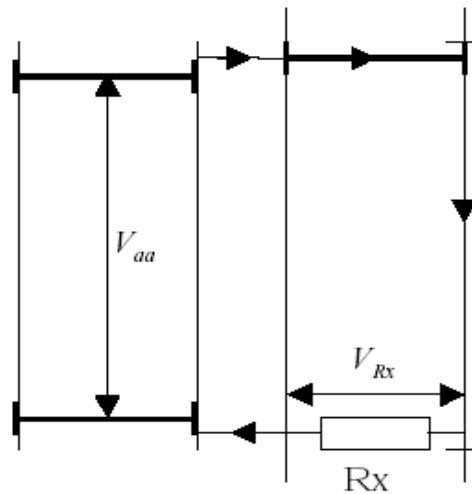


Figure 12: Wrong side failure caused by axle-to-axle voltage, single rail track circuit

In the limiting case the maximum voltage allowable at the relay when there is a train on the track circuit is the drop-away voltage. In the worst case the axle to axle voltage of the train on the adjacent track is presented directly across the relay. If an interference budget of one third is allowed for the train this leads to the maximum permissible axle to axle voltage being set at one third of the relay drop away voltage.

$$V_{aa \max} = \frac{V_{da}}{3} \quad (7.13)$$

If the same relay as used in section 6.2.5 is taken as an example, the drop away voltage is 0.45 V leading to a permissible axle to axle voltage of 0.15 V.

8. Train Data For Analysis

8.1 Normal train condition

The aim of the analysis of a train operating under normal conditions is to establish the maximum steady state interference current and transient interference current. Where normal conditions is taken to mean the all modes of operation that a train can be expected to experience i.e. braking; accelerating; coasting; auxiliaries on; auxiliaries off; subsystems shut down or isolated etc.

The transient current analysis must also consider all possible events, which could give rise to transients. Table 3 gives some conditions which must be taken into account as they are known to give rise to transients, other events may also give rise transients and the manufacturer must ensure that a process is in place so that all such events are identified and considered.

No.	Event
1	Start up, loaded
2	Acceleration, loaded
3	Circuit breaker trip
4	Circuit breaker closure
5	Circuit breaker auto-reclosure
6	Shoe bounce
7	Gapping
8	Bad adhesion condition / Wheel slip
9	Rheostatic braking
10	Regenerative braking
11	Bad current collection conditions / Contact problems
12	Power on / power off switching

Table 3: Events used in the transient analysis.

8.2 Failed train condition

Train failures, which may increase the EMI with respect to 50 Hz single rail track circuits, must be considered and faults, which may result in EMI, identified. The methodology to be employed to identify such faults is beyond the scope of this document. However when such faults are identified their effect on steady state and the transient performance of the train has to be assessed. The effects of faults on the EMI profile of the train may either be measured by enacting the fault under controlled conditions or by modelling. In either event the track circuit model shall be used to assess the effect of the resulting EMI on the operation of 50 Hz single rail track circuits. An assessment is then undertaken to judge whether the performance of the train with a particular fault coupled with the likelihood of that fault occurring represent an unacceptable risk to the track circuits.

9. Methodology For Demonstration Of Compliance

The longest track circuit on the route of interest must be determined by recourse to the RAR (Railtrack Asset Register) or other source. Any 'long' track circuits that are found will require examination on a case by case basis.

It must be confirmed that the track circuits are installed in single rail mode only.

The rail-break figures and times at risk are indicative only and should be replaced by the latest available data and/or statements from Railtrack [10].

9.1 Normal train - normal infrastructure

The methodologies described in sections 5.2.5 should be used to determine a steady state level for the track circuits of interest which can then be compared to the train generated interference under normal steady state operations this limit should not be exceeded.

The methodology described in section 6 should be used to evaluate the effect of transient interference generated by the 'new' train in comparison to the 'old' trains.

9.2 Faulty train - normal infrastructure

From the failure analysis carried out the rate at which the train exceed the limit calculated for normal infrastructure as in section 9.1 should be determined. The frequency of occurrence, duration and magnitude of the exceedance should be taken into account when constructing mitigating arguments.

9.3 Normal train - faulty infrastructure

The faulty infrastructure scenario as described in section 5.2.6 is used to determine a permissible current for a given fault scenario. This can then be apportioned down by an allowance to determine a train current limit. In the case of faulty infrastructure a reduced allowance for interference from other sources can be applied, in this case a factor of 2 can be used. The train current limit so calculated can then be compared to the train data. The rate at which the train exceeds this limit in conjunction with the occurrence of this failure mode can then be calculated. These events should be quantified in terms of the rate at which they occur and the overall rate of potential WSFs is determined by combining them with track failure data.

This can be done in the following manner.

For the case of a broken rail occurring on a particular track circuit. The total number of rail breaks per year is 680 (ref.10) per 20,000 track miles and the time at risk is 12 hours taken as the average time for a rail break to be discovered. The number of railbreaks per year for the section of railway of interest must be confirmed by recourse to latest available figures or statements available from Railtrack. The probability of a track circuit having a broken traction rail at any one time is:

Probability of track circuit having broken rail (Ptcbr) =
No of breaks/track miles * TC length(miles)*(time at risk/no of hours per year)

The predicted rate for a train that is standing on a track circuit exceeding the susceptibility level and the traction rail being broken is therefore:

Ptcbr * Exceedance rate (per hour)

This will then give the failure rate in exceedances per hour for normal train and failed infrastructure.

9.4 Faulty train - faulty infrastructure

In this case the steady state train limits and the probability of a track circuit having a broken rail (Ptcbr) are determined in the same way as in section 9.3. The exceedance rate for train failures is predicted by the failure analysis. This failure rate is then used in conjunction with the probability of a track circuit having a broken rail to determine the overall exceedance rate for this case.

9.5 Axle to axle voltages

The axle to axle voltages should be compliant with a limit calculated using the methodology described in section 7.

APPENDIX A: PARAMETERS

Parameter values used

- VTI relay, applying the drop away voltage, i.e. $V_{Rx} = 0.45 \text{ V}$,
- Relay 50 Hz impedance: $Z_{Rx} = 3.36 \Omega + j 0.228 \Omega$ [5]
- Per length rail 50 Hz impedance: $Z'_{Sig} = Z'_{Tr} = (1.694 \cdot 10^{-4} + j 3.48 \cdot 10^{-4}) \Omega/\text{m}$ [5]
- Train shunt resistance: $R_{Sh} = R_{Sh1} = R_{Sh2} = 100 \mu\Omega$

N.B. this is small compared to other elements in the analysis and is ignored in the final steady state analysis

- Track circuit length and train distance from relay end: distance = 200 m
- Tail resistance of the track circuit relay: $R_{Tail} = 0$

APPENDIX B: ASSUMPTIONS

- 50 Hz single rail track circuits are correctly installed and maintained in accordance with the Track Circuit Manual [1].
- The length for assessment of the 50 Hz single rail track circuit is assumed to be 200 m.
- No track repeat relays (TPR) are fitted.
- To consider a normal (unfailed) pessimistic track circuit, all traction current is assumed to flow via the traction rail in one direction.
- Absence of the Westalite stabilising rectifier

Infrastructure failure modes

The failure mode considered is that of a broken rail, which has the affect of extending the current return path via cross bonds and adjacent track.

APPENDIX C: HAZARDS

Interference current in traction return current

This hazard is considered in this assessment.

Longitudinal voltage

This hazard is considered in this assessment.

Direct induction into track mounted or trackside components

This mechanism is not considered to be a significant risk to the correct operation to 50Hz track circuits, however any assessment should consider the magnitude of magnetic fields likely to be produced and their effect on 50 Hz track circuit operation.

APPENDIX D: MITIGATING FACTORS

Pessimistic assumptions

Use of the drop-away voltage of the 50 Hz vane relay.

Maximum track circuit length 200m

Train located at far end of track circuit.

Broken rail is worst case location to set up and undetected failure.

Zero train length for line current limits.

All of the return current flows in the rails.

Assumptions/variations covered by safety margin

Normal variation of track circuit receiver performance.

Background/coincident sources of interference covered by safety margin

Other rolling stock.

Other interference sources such as track feeds.

APPENDIX E: SOURCES OF TECHNICAL WORK

The technical work contained in this document is taken from the Class 365 EMC safety case, produced by AEA Technology Rail, this work was reviewed by WS Atkins Rail as the ISA on behalf of Eversholt Leasing, later HSBC Rail. Much of the work was taken from the “50 Hz Track Circuit Study” undertaken for Eversholt Leasing and broadly connected with Networker trains, but not explicitly connected to the Safety Assessment. Additional technical work has also been taken from work carried out for the ALSTOM ‘Juniper’ trains. This work was undertaken by AEA Technology Rail, the ISA was WS Atkins.

APPENDIX F: RAIL AND RELAY IMPEDANCE FREQUENCY CHARACTERISTIC

F.1 Rail Impedance

The values of rail impedance are taken from reference [5] and are as follows

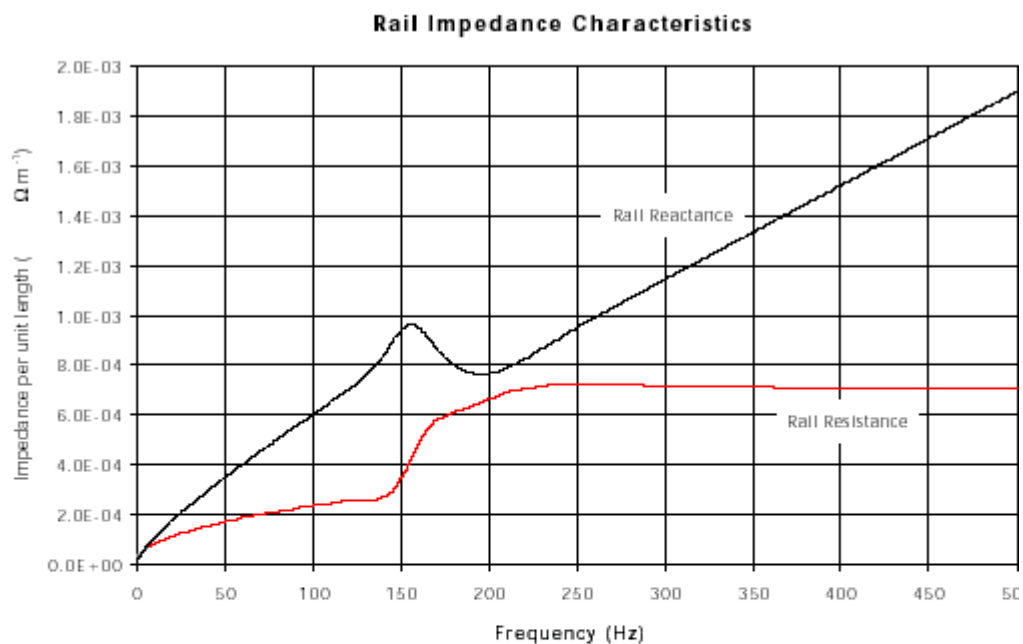


Figure F.1.1: Rail impedance characteristic.

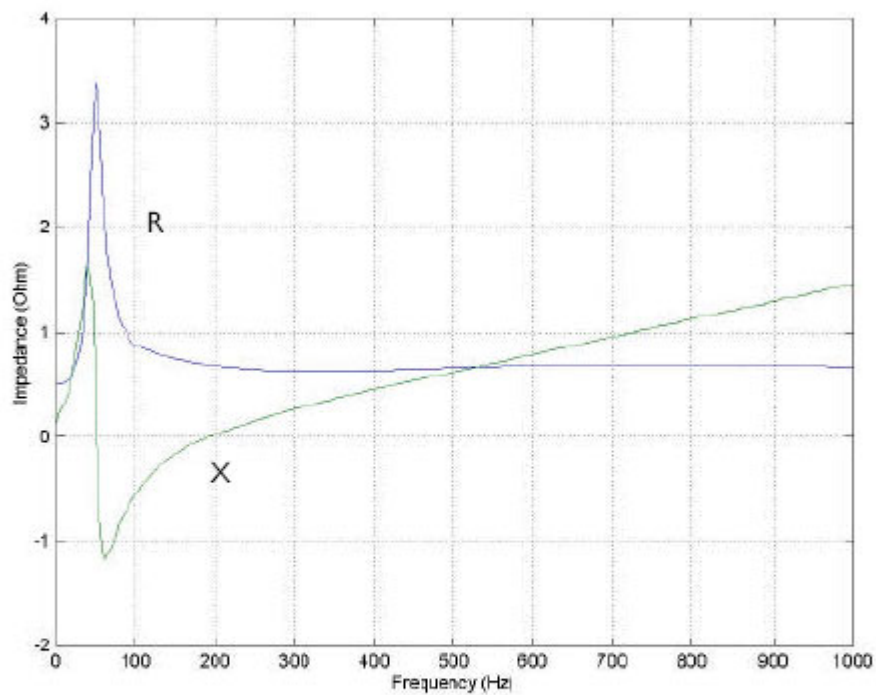
Frequency (Hz)	Impedance per km (Ω)	
	Model	'Train detection' [3]
0	0.04	0.035
50	0.39	0.3
400	1.68	2.5
2000	7.57	10

Table F.1.1 Comparison of rail impedance magnitudes.

F.2 Relay impedance frequency characteristic

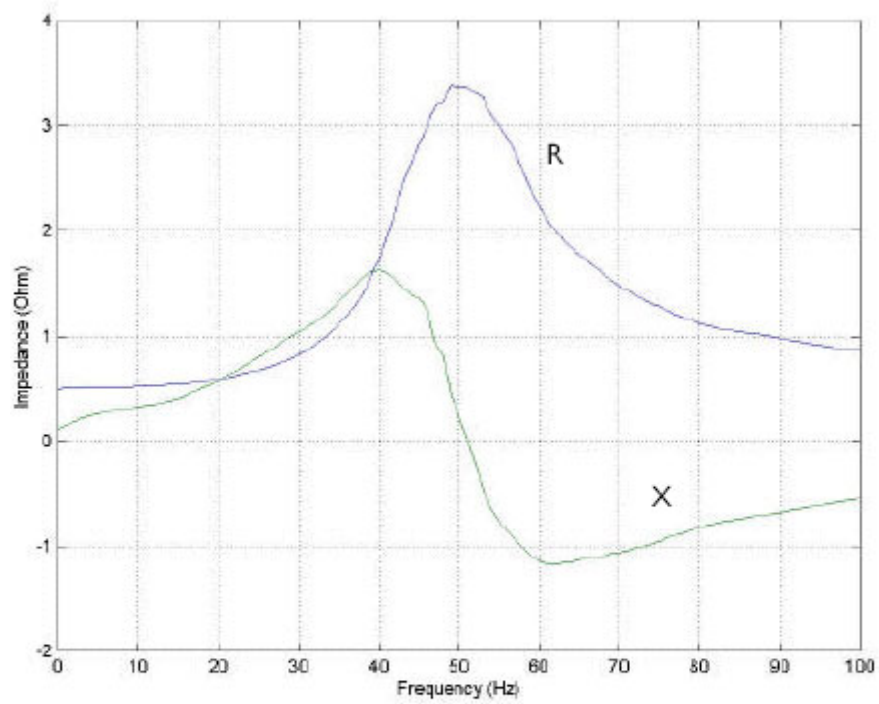
Tests carried out on the VTI track circuit relay showed that mutual coupling between the control coil and the local coil can be neglected [7]. As a result of this,

the relay control coil can be measured independently of the local coil connection with the tuning capacitor connected to the control coil secondary winding. Fig. 2 shows the impedance-frequency characteristic of the vane relay control coil, and table 2 gives the resulting look up tables.



$$0 \text{ Hz} < f < 1000 \text{ Hz}$$

Fig. F.2.1 Measured VTI vane relay control coil impedance



$$0 \text{ Hz} < f < 100 \text{ Hz}$$

Fig. F.2.1 Measured VTI vane relay control coil impedance

f / Hz	R / Ω	X / Ω
0	0.5	0.11
20	0.59	0.58
30	0.82	1.04
35	1.11	1.32
40	1.73	1.63
45	2.83	1.36
50	3.36	0.228
55	3.0	-0.761
60	2.23	-1.14
65	1.78	-1.13
70	1.48	-1.06
75	1.28	-0.95
80	1.12	-0.82
90	0.98	-0.68
100	0.88	-0.55
150	0.746	-0.162
200	0.68	0.0201
300	0.627	0.264
400	0.633	0.454
500	0.66	0.62
700	0.685	0.956
1000	0.67	1.46

Table F.2.1 Relay Impedance VT1. From reference 5.

RAILTRACK

BRIEFING NOTE

RT/E/C/50001 to RT/E/C/50019

Issue 1

Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure

Issue Date: February 2003

Background / Synopsis

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

Authors of Route Acceptance Safety Cases.

Name and Title of Author

David Wilkinson
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